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RESEARCH ARTICLE

Fine-Grained Scheduling Strategies and Optimization of Wind–Solar–Storage Powered Data Center Microgrids

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ABSTRACT This study presents a comprehensive collaborative optimization strategy for the efficient allocation of wind, solar, and energy storage capacities in data center microgrids. The proposed strategy systematically integrates multiple decision-making factors, including the intensity of renewable power generation, real-time electricity prices, and task completion times, with the dual objectives of minimizing total operational costs and carbon emissions while maximizing user satisfaction. To address the heterogeneous nature of data center workloads, the strategy differentiates between interactive and batch tasks, leveraging their respective scheduling flexibilities. Thirteen distinct time window models are employed to fully exploit temporal flexibility, enabling tasks to be adaptively shifted in response to renewable generation and price fluctuations. The optimization framework ensures that renewable energy is prioritized while meeting system reliability constraints. Simulation results under various capacity configurations validate that the proposed approach significantly improves renewable energy utilization, reduces electricity expenses, and enhances user-perceived service quality. By offering both a theoretical framework and practical implementation guidelines, this research provides valuable insights into capacity planning, operational scheduling, and sustainable management of next-generation data center microgrids.

INDEX TERMS Data center microgrid, wind–solar–storage capacity, scheduling strategy, user satisfaction.

I. INTRODUCTION

The rising electricity consumption of data centres has made their green transformation an urgent task in the pursuit of global sustainability goals [1]. In recent years, microgrid architectures based on renewable energy sources—such as wind, solar, and energy storage—have emerged as effective solutions for improving energy efficiency and reducing carbon emissions in data centres [2], [3]. International initiatives, including the United Nations Framework Convention on Climate Change and the European Green Deal, have further accelerated the low-carbon transition of the data centre industry [4]. However, the inherent intermittency and uncertainty of wind and solar power generation pose significant challenges. Determining how to optimally size wind turbines, photovoltaics, and energy storage systems, while leveraging the temporal and spatial flexibility of data centre workloads, is key to ensuring system stability and operational efficiency [5], [6], [7], [8].

In response, numerous studies have been conducted. Liu et al. [9] proposed a deep reinforcement learning approach for energy-aware scheduling to obtain an optimal online policy. Real-world case studies show it effectively

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TABLE 1. Variable names and symbols.

Name	Svmbolsymbol	Name	Symbolsymbol
Capital Cost	f	Wind turbine capital cost	C_1
PV capital cost	C_2	Energy storage capital cost	C_3
Wind turbine maintenance cost	C_4	PV maintenance cost	C_5
Energy storage maintenance cost	C_6	Electricity purchase cost	C_7
Electricity selling revenue	C_8	Wind annualized cost coefficient	θ_w
PV annualized cost coefficient	θ_{pv}	Wind unit installation cost	c_w
Wind discount rate	r_0	Wind system lifetime	Y_w
Total experiment duration	T	PV unit installation cost	c_{pv}
PV discount rate	r_1	PV system lifetime	Y_{pv}
Energy storage capacity	y_e	Energy storage capacity installation cost	c_e^{ins}
Energy storage discount rate	r_2	Energy storage lifetime	Ye
Wind Turbine Maintenance Cost	c_{wt}^{mai}	PV Maintenance Cost	c_{pv}^{mai}
Energy Storage Maintenance Cost	c_e^{mai}	Electricity purchase price	P_1
Power Purchased from the Grid	P_{line}^t	Sampling interval	$\triangle T$
Electricity selling price	P_2	Power Exported to the Grid	P_{ex}^t
Total carbon emissions	C_f	Carbon intensity of the power grid	C_f^g
Wind carbon intensity	C_f^{wt}	PV carbon intensity	C_f^{pv}
Overall user satisfaction	w	Batch processing user i satisfaction	S_i
Interactive user i satisfaction	F_i	Task completion time	T_{com}
Task delay time	T_{del}	Rated wind power	P_{wr}
Rated PV power	P_{pvr}	Wind power at time t	P_w^t
PV power at time t	P_{pv}^t	Load power at time t	P_{mom}^t
Charging power at time t	P_{chr}^t	Discharging power at time	P_{dis}^t
Min charging power	$P_{chr.\min}$	Max charging power	$P_{chr.\max}$
Min discharging power	$P_{dis.\min}$	Max discharging power	$P_{dis.\max}$
Charging/discharging state	X/Y	Min SOC (storage)	$S_{OC.\min}$
Max SOC (storage)	$S_{OC.\max}$	SOC at time t	S_{OC}^t
Max wind power output	$P_{w.\max}$	Max PV power output	$P_{pv.\max}$
Number of servers at time t	Q_t	Total number of servers	$Q_{\max}$
Net load power	P_N	Loss of load power	P_L
Ideal max charging power	$P_{chr,soc}$	Ideal max discharging power	$P_{dis,soc}$
Charging efficiency	γ_{chr}	Discharging efficiency	γ_{dis}
Number of tasks at time t	T_{ask}	Rated Power of Energy Storage	E_{nergy}
Energy storage annualized cost coefficient	θe		

reduces data center carbon footprint and energy cost while ensuring cloud service quality. Al Kez et al. [10] proposed a conceptual framework that characterizes Internet data centers as smart and responsive assets capable of participating in

demand response programs. By aligning their scheduled operations with grid dynamics, such assets can significantly contribute to the integration of high shares of variable renewable energy. Li et al. [11] investigated the problem of temporal load flexibility scheduling in data centers under delay-tolerant quality of service requirements. By leveraging the inherent delay tolerance of workloads and exploiting electricity price variations across different time periods, they enabled time-shifted workload migration within the data center to reduce operating costs. Similarly, major operators such as Google and Facebook have adopted workload migration and energy forecasting to optimize renewable energy utilization [12]. However, these studies predominantly focus on generation-side modelling or batch tasks, lacking a systematic exploration of the coordination mechanisms between interactive and batch tasks. In addition, existing studies lack comprehensive modeling of multi-scenario dynamic interactions between energy storage and the grid, and rarely account for energy storage degradation [13], [14], [15].

Further research has shown that incorporating load flexibility is an effective approach to enhancing scheduling performance. Meng et al. [16] introduced a time-utility function to capture the time-sensitive value of tasks and proposed a decomposition-based scheduling algorithm that prioritizes resource allocation, significantly improving service quality under heterogeneous workloads. Xiang et al. [17] modeled batch workload response across multiple time-sensitive windows to enable spatiotemporal migration, forming a demand response mechanism that reduces operational costs while maintaining user satisfaction. Yu et al. [18] implemented incentive mechanisms to encourage users to actively adapt load profiles across time and space, thereby increasing renewable energy utilization and reducing emissions without compromising task completion. More recently, reinforcement learning and deep neural networks have been employed to enhance scheduling adaptability under environmental uncertainties [19], [20]. However, current studies largely concentrate on flexible scheduling of batch tasks, often neglecting the modelling and optimisation of interactive workloads. Moreover, the limited number of time windows used in some studies restricts the ability to fully represent temporal flexibility, constraining the generalisability and real-world applicability of the proposed strategies. Consequently, developing a scheduling approach that integrates the characteristics of both interactive and batch workloads—while offering fine-grained temporal responsiveness—remains a pressing research challenge [21].

To address the aforementioned limitations, this paper proposes a multi-objective collaborative optimization scheduling and capacity configuration strategy for multi-source heterogeneous data center microgrids, with innovations in three key aspects:

1. Task-type collaboration: A joint scheduling framework integrating interactive and batch workloads, incorporating task timeout handling and energy storage protection mechanisms, enhancing task completion and system stability. This directly addresses the lack of systematic coordination in prior studies [22], [23].
2. *WSETSS* joint scheduling strategy: A multi-dimensional coupling scheduling approach based on renewable generation, electricity prices, and task completion times. It captures workload flexibility through multiple time windows and evaluates diverse capacity configurations. Unlike existing reinforcement learning or hybrid methods, *WSETSS* requires no large-scale training or iterative optimization, providing a more interpretable, deployable, and practical solution [24], [25], [26], [27].
3. Dynamic energy storage–grid coordination model: Sixteen typical energy interaction scenarios are constructed, including energy storage degradation constraints, to realistically simulate the dynamic coupling among energy storage, the grid, and data centers. This addresses gaps in prior works that overlooked multi-scenario interactions and storage degradation modeling [28], [29].

The main contributions of this work are summarized as follows:

1. Multi-task collaborative scheduling mechanism: Integrates interactive and batch tasks with task timeout handling and storage protection, improving system coordination and operational stability.
2. *WSETSS* joint optimization strategy: Exploits task flexibility through multi-dimensional coupling scheduling, enhancing scheduling efficiency and renewable energy utilization, clearly distinguishing it from existing RL and hybrid approaches.
3. Dynamic energy storage–grid coordination model: Incorporates multiple energy interaction scenarios and storage degradation constraints to enable more realistic energy flow modeling and control.

This work provides a practical framework combining interactive/batch scheduling, WSETSS optimization, and dynamic storage–grid coordination, introducing innovations in task flexibility, storage degradation, and multi-energy scheduling, clearly distinct from existing RL and hybrid methods.

In view of the extensive use of mathematical symbols in this study, the relevant notations are compiled in Table 1 for clarity and ease of reference.

II. OPTIMAL CAPACITY CONFIGURATION MODEL OF THE DATA CENTER MICROGRID

In the optimal capacity configuration of data center microgrids, improving operational efficiency and effectively controlling investment costs are two critical objectives. This study focuses on three key dimensions—economic

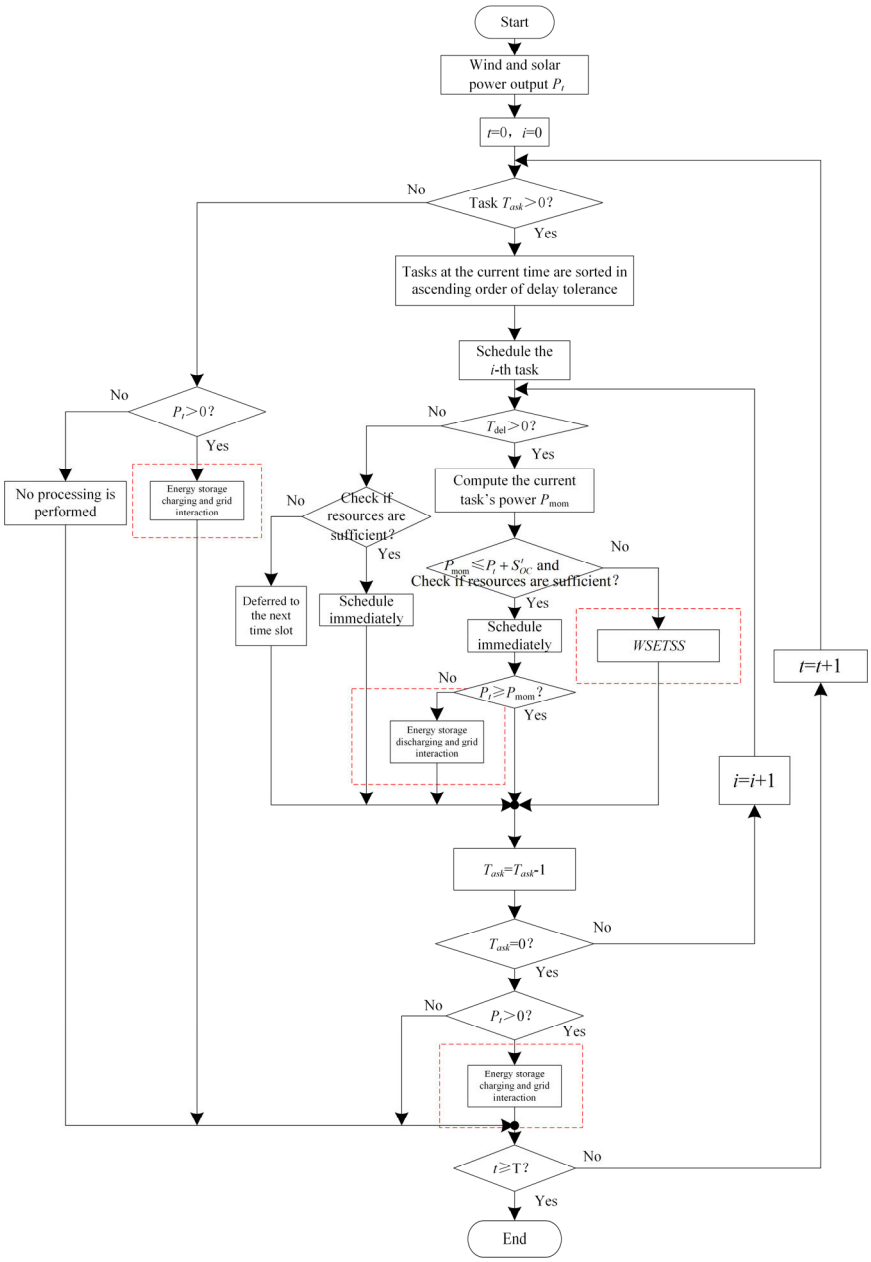


FIGURE 1. The dispatching and operation strategy of the data center microgrid.

performance, carbon reduction, and user satisfaction—to conduct an in-depth investigation. Accordingly, three objective functions are formulated to minimize total cost, reduce carbon emissions, and maximize user satisfaction in the data center microgrid system. The optimization model is presented in Equation (1) and Equation (2).

$$\begin{cases} \min f = C_1 + C_2 + C_3 + C_4 + C_5 + C_6 + C_7 - C_8 \\ \min C_f = (P^t_{line} C^g_f - P^t_w C^{wt}_f - P^t_{pv} C^{pv}_f) T \\ \max w = \sum_{i=1}^{m} (S_i + F_i) \end{cases} \tag{1}$$

$$\text{s.t.} \begin{cases} \begin{cases} P^t_w + P^t_{pv} = P^t_{mom} + P^t_{chr} + P^t_{ex}, P^t_w + P^t_{pv} \geq P^t_{mom} \\ P^t_w + P^t_{pv} + P^t_{dis} + P^t_{line} = P^t_{mom}, P^t_w + P^t_{pv} < P^t_{mom} \end{cases} \\ P_{chr.\min} \leq p^t_{chr} \leq P_{chr.\max} \\ P_{dis.\min} \leq p^t_{dis} \leq P_{dis.\max} \\ X = 1 - Y \\ S_{OC.\min} \leq S^t_{OC} \leq S_{OC.\max} \\ 0 \leq P^t_w \leq P_{w.\max} \\ 0 \leq P^t_{pv} \leq P_{pv.\max} \\ Q_t \leq Q_{\max} \end{cases} \tag{2}$$

In Equation (1), f represents the operating cost of the data center microgrid; C_1, C_2, and C_3 denote the installation costs of wind turbines, photovoltaics, and energy storage systems, respectively; C_4, C_5, and C_6 correspond to their maintenance costs; C_7 represents the cost of electricity purchased from the grid; C_8 is the revenue from selling electricity back to the grid. C_f indicates the total carbon emissions, while C_f^g, C_f^{wt}, and C_f^{pv} refer to the carbon intensities of grid electricity, wind power, and solar power, respectively. w denotes user satisfaction, with S_i and F_i representing the satisfaction levels of batch user i and interactive user i, respectively.

In Equation (2), the first two rows represent power balance constraints; rows 3–6 define the charging and discharging constraints of the energy storage system; row 7 constrains the output power of wind turbines; row 8 constrains the output of photovoltaics; and row 9 represents the server quantity constraint.

III. MULTI-DIMENSIONAL COLLABORATIVE OPTIMIZATION SCHEDULING STRATEGY FOR DATA CENTER MICROGRIDS

A. CHARACTERISTICS OF THE DATA CENTER MICROGRID SCHEDULING STRATEGY

1) DESIGN OF A TASK RESIDUAL JUDGMENT MECHANISM

To address the issue that wind and solar renewable energy remaining available after load processing may lead to frequent short-term charge-discharge cycles that can reduce energy storage lifespan, a task residual judgment mechanism is designed. Upon completion of the current task, this mechanism assesses whether there are pending tasks. If so, the remaining renewable energy is preferentially allocated to subsequent tasks; only when no residual tasks remain will the system initiate energy storage charging or interact with the grid. This approach effectively prolongs the service life of the energy storage system while optimizing the overall energy utilization pathway.

2) INCORPORATION OF A TASK TIMEOUT POSTPONEMENT MECHANISM

A task timeout postponement mechanism is applied to interactive and batch loads, handling small-scale timeout events through deferred processing to enhance system robustness and stability.

B. SCHEDULING STRATEGY PROCESS

As illustrated in Fig.1, the system operates on a rolling basis with a 5-minute interval, during which wind and solar generation data as well as load demand information are collected. For each time slot t the system first calculates the available power based on the wind and solar output at the current time, and then determines whether there are any pending tasks [30]. The scheduling process is categorized into two main scenarios, as detailed below:

1) **Case of No Pending Tasks:**

If there are no pending tasks at the current time ($T_{ask} = 0$), the system further evaluates the wind and solar power output (P_t). If $P_t > 0$, the surplus energy is first used to charge the energy storage system. The system then makes a dynamic decision on whether to interact with the grid based on the microgrid's current energy status. If $P_t = 0$, no energy scheduling operation is performed, and the system proceeds to the next step. Subsequently, the system checks whether the current time t has reached the maximum scheduling horizon T. If so, the scheduling loop terminates; otherwise, the system advances to the next time slot and continues the scheduling process.

2) **Case with Pending Tasks:**

If $T_{ask} > 0$, the system first sorts all tasks in ascending order based on their allowable delay time. Tasks are then processed sequentially. Depending on task type and scheduling conditions, the logic is further divided into the following three cases:

(1) **Interactive Tasks:**

If a task's delay tolerance is $T_{\text{del}} = 0$, the system evaluates whether sufficient computational resources are available at the current time. If available, the task is executed immediately. Otherwise, the task is deferred to the next time slot and is rescheduled until adequate resources become available. After task execution, the system checks whether any unprocessed tasks remain at the current time. If $T_{ask} > 0$, the scheduling loop continues; if $T_{ask} = 0$, the system evaluates whether surplus power P_t remains. If $P_t > 0$, the energy storage system is charged and grid interaction is considered based on the microgrid's energy status; if $P_t = 0$, no energy scheduling is performed. Finally, the system checks whether time t has reached T to determine whether to terminate the loop or proceed to the next cycle.

(2) **Batch Tasks (Executable at Current Time):**

The required power P_{mom} for the current task is calculated, and the system checks whether both the power requirement ($P_{\text{mom}} \leq P_t + S_{OC}^t$) and computational resource constraints are satisfied. If so, the task is executed immediately. If renewable output is insufficient during execution, the energy storage system compensates for the shortfall, and the system interacts with the grid based on the microgrid's operational status. If renewable generation is sufficient, the next task is scheduled directly. Subsequently, the system evaluates whether there are remaining tasks ($T_{ask} > 0$), and performs energy dispatch based on the remaining P_t. Finally, time t is compared with the upper bound T to determine whether to continue the scheduling process.

(3) **Batch Tasks (Not Executable at Current Time):**

If either the power or computational constraints are not met, the *WSETSS* scheduling strategy is applied to handle the task. Upon completion, the system again evaluates remaining tasks and dispatches surplus energy for storage charging or grid interaction. The final step is to assess whether the scheduling horizon T has been reached.

TABLE 2. Comparison of five scheduling strategies.

Scheduling Strategy	Priority Criterion	Expected Advantage	Potential Limitation
Strategy A	Periods with high renewable output > periods with low price	Minimizes carbon emissions	Low user satisfaction
Strategy B	Next-period renewable output > periods with low price	High user satisfaction	Higher operational cost
Strategy C	Periods with high renewable output > periods with fastest task completion	Minimizes carbon emissions	Low user satisfaction
Strategy D	Next-period renewable output > periods with fastest task completion	High user satisfaction	Higher operational cost
Strategy E	Periods with low price	Reduces operational cost, initially high user satisfaction	Higher carbon emissions

IV. WSETSS SCHEDULING STRATEGY

WSETSS is a scheduling strategy designed for batch processing loads, which comprehensively considers three key factors: wind and solar power output, electricity price, and the shortest task completion time. By systematicall prioritizing these indicators and eliminating redundant strategies, five representative scheduling schemes are obtained to enable more refined and efficient management of batch workloads [31]. Its characteristics are summarized in Table 2.

A. WIND AND SOLAR OUTPUT (MAXIMUM PRIORITY) + MINIMUM ELECTRICITY

As illustrated in Fig.2, this strategy adopts a two-stage scheduling mechanism. In the first stage, tasks are assigned to the time slot with the highest combined wind and solar power output. If the available renewable power Pmax at that moment satisfies the task's power requirement Pmom $\leq$ Pmax, and computing resources are sufficient, the task is immediately scheduled at that moment. If either condition is not satisfied, the task is sequentially scheduled to subsequent time slots with lower wind and solar output (e.g., $P_{\max-1}$, $P_{\max-2}$), until all eligible time windows have been exhausted.

If scheduling cannot be completed in the first stage, the system switches to the second stage, where task allocation is based on the lowest electricity price. The system locates the time slot with the minimum price $P_{1.\min}$ and attempts to schedule the task. If computing resources are insufficient at that moment, it continues to search time slots with the next lowest electricity prices (e.g., $P_{1.\min-1}$, $P_{1.\min-2}$) until a feasible option is found. If all candidate slots fail to meet the scheduling criteria, the task is deferred to execute after its maximum allowable delay time T_{del}.

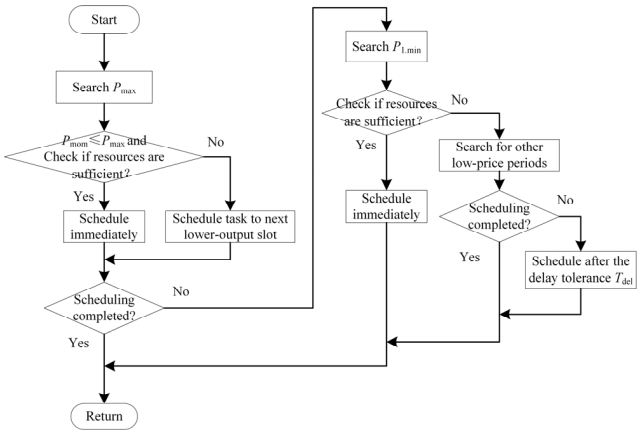


FIGURE 2. Wind and solar output (maximum priority) + minimum electricity price.

B. WIND AND SOLAR OUTPUT (SEQUENTIAL) + MINIMUM ELECTRICITY PRICE

As shown in Fig.3, this strategy adopts a two-stage scheduling mechanism. In the first stage, wind and solar power outputs are prioritized in chronological order. The scheduling system sequentially examines the available renewable power starting from time step $t + 1$. If, at any time slot, the task's power requirement $P_{\text{mom}} \leq P_{t+1}$ is met and sufficient computing resources are available, the task is immediately scheduled at that moment. If either condition is not satisfied, the system continues searching the subsequent time slots until all available scheduling windows have been traversed.

If no feasible scheduling decision can be made in the first stage, the strategy switches to the lowest electricity price–based mechanism described in A. The system searches time slots in ascending order of electricity price and schedules the task based on resource availability. If the task still cannot be scheduled, it will be deferred to execute after its maximum allowable delay time T_{del}.

C. WIND AND SOLAR OUTPUT (MAXIMUM PRIORITY) + SHORTEST COMPLETION TIME

As shown in Fig. 4, this strategy begins by scheduling tasks during the period of highest renewable power output,

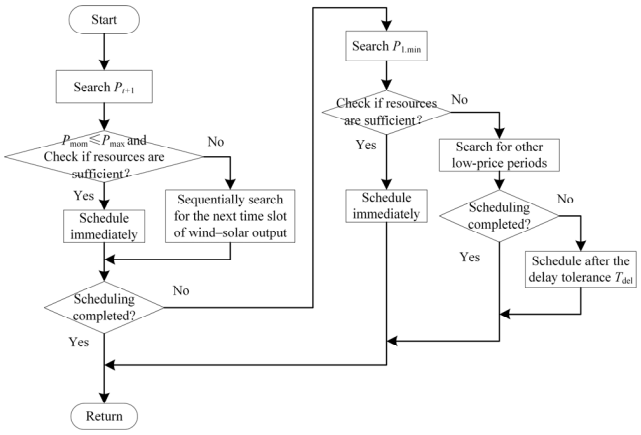


FIGURE 3. Wind and solar output (sequential) + minimum electricity price.

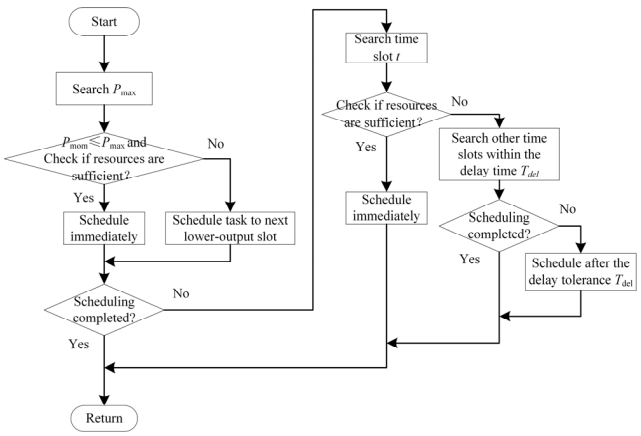


FIGURE 4. Wind and solar output (maximum priority) + shortest completion time.

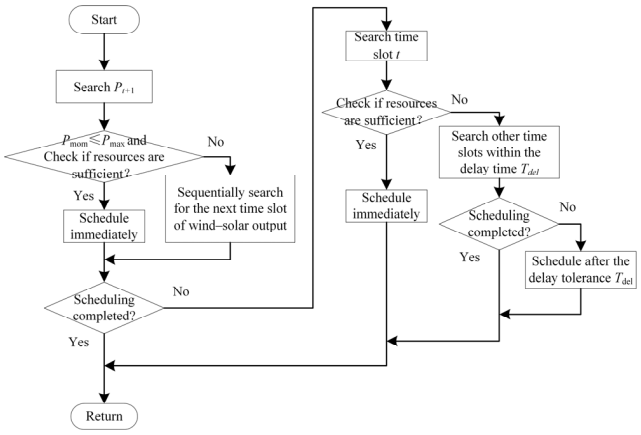


FIGURE 5. Wind and solar output (sequential) + shortest completion time.

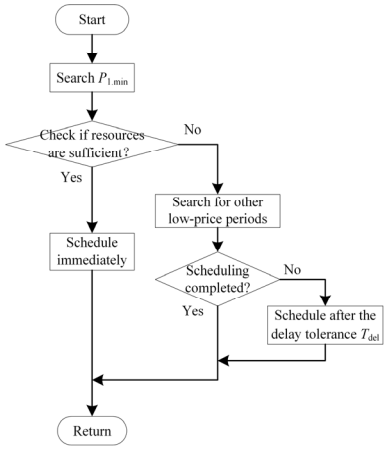


FIGURE 6. Minimum electricity price.

following the same logic as A. If power and computing resource conditions are met, the task is scheduled immediately.

If the first stage fails, the strategy switches to a shortest completion time policy. From the task's arrival time onward, time slots are examined in chronological order to identify one with sufficient computing resources. If none is found, the task is deferred to execute after T_{del}.

D. WIND AND SOLAR OUTPUT (SEQUENTIAL) + SHORTEST COMPLETION TIME

This strategy, as illustrated in Fig.5, combines sequential renewable utilization with the shortest completion time principle. In the first stage, renewable output is checked sequentially from the current time. If the output at a time slot satisfies P_{mom} and resources are available, the task is scheduled accordingly.

If no suitable slot is found, the system initializes the current time ttt and applies the shortest completion time policy (as described in C). It then searches for a time slot from ttt onward that satisfies computing resource constraints. If no feasible slot is available, the task is scheduled after T_{del}.

E. MINIMUM ELECTRICITY PRICE

As shown in Fig.6, this strategy employs a single-stage scheduling mechanism that focuses exclusively on minimizing electricity costs, following the same logic as the second stage of A. Starting from the current time, the system sequentially searches available time slots in ascending order of electricity price. If sufficient computing resources are available in a given slot, the task is scheduled accordingly; otherwise, it is deferred until its maximum allowable delay time T_{del}.

V. DYNAMIC ENERGY STORAGE AND GRID COORDINATED CONTROL STRATEGY

In the process of charging and discharging the energy storage system, eleven key parameters are introduced, namely P_N, P_L, $P_{chr.\min}$, $P_{chr.\max}$, $P_{dis.\min}$, $P_{dis.\max}$, $S_{OC.\min}$, $S_{OC.\max}$, $St\ OC$, $P_{chr,soc}$, and $P_{dis,soc}$, with their corresponding expressions provided in Equation (3). By analyzing the interrelationships among these parameters, the operation states

of the energy storage system are classified into sixteen representative scenarios. This classification enables more accurate simulation of the system's dynamic response under real-world charging, discharging, and grid-interaction conditions [32].

$$\begin{cases} P_N = P_w^t + P_{pv}^t - P_{\text{mom}}^t \\ P_L = P_{\text{mom}}^t - P_w^t - P_{pv}^t \\ P_{chr,soc} = \frac{S_{OC.\max} - S_{OC}^t}{\gamma_{chr}\Delta T} E_{\text{nergy}} \\ P_{dis,soc} = \frac{(S_{OC}^t - S_{OC.\min})\gamma_{dis}}{\Delta T} E_{\text{nergy}} \end{cases} \tag{3}$$

Specifically, P_N denotes the net load power, and P_L represents the loss of load power. $P_{chr,\text{soc}}$ and $P_{dis,\text{soc}}$ refer to the maximum charging and discharging power under ideal conditions, respectively, and are used to characterize the upper bounds of the energy storage system's actual charging and discharging capabilities under state-constrained conditions.

1) Case(1)

When $P_N < P_{chr,\min}$, P_N cannot be charged to the energy storage and will be charged to electricity The network sells electricity.

2) Case(2)

When $P_{chr,\min} < P_N < P_{chr,\max}$ and $P_{chr,soc} < P_{chr,\min}$, P_N is unable to charge the energy storage and will sell electricity to the grid.

3) Case(3)

When $P_{chr,\min} < P_N < P_{chr,\max}$, $P_{chr,\min} < P_{chr,soc} < P_{chr,\max}$ and $P_{chr,soc} < P_N$, the energy storage system can be charged using the available power. Since $P_{chr,soc} < P_N$, the storage charges at $P_{chr,soc}$ until it reaches full capacity, and the remaining power is exported to the grid.

4) Case(4)

When $P_{chr,\min} < P_N < P_{chr,\max}$, $P_{chr,\min} < P_{chr,soc} < P_{chr,\max}$, and $P_{chr,\max} < P_{chr,soc}$, the energy storage system can be charged. Since $P_N < P_{chr,soc}$, the storage charges at P_N, but does not reach full capacity, and no power is exported to the grid.

5) Case(5)

When $P_{chr,\min} < P_N < P_{chr,\max}$ and P$P_{chr,\max} < P_{chr,so}$, the energy storage system can be charged. Since $P_{chr,\max} < P_{chr,soc}$, the storage charges at P_N, but does not reach full capacity, and no power is exported to the grid.

6) Case(6)

When $P_{chr,\max} < P_N$ and $P_{chr,soc} < P_{chr,\min}$, the energy storage system cannot be charged, and the excess power is sold to the grid.

7) Case(7)

When $P_{chr,\max} < P_N$ and P$P_{chr,\min} < P_{chr,soc} < P_{chr,\max}$, the energy storage system can be charged. The storage charges at $P_{chr,soc}$ until it reaches full capacity, and any surplus power is sold to the grid.

8) Case(8)

When $P_{chr,\max} < P_N$ and $P_{chr,\max} < P_{chr,soc}$, the energy storage system charges at its maximum allowable rate $P_{chr,\max}$ but cannot be fully charged. The remaining power P_N- $P_{chr,\max}$ is exported to the grid.

9) Case(9)

When $P_L < P_{dis,\min}$, the energy storage cannot be discharged, and the required power is purchased from the grid.

10) Case(10)

When $P_{dis,\min} < P_L < P_{dis,\max}$ and $P_{dis,soc} < P_{dis,\min}$, the energy storage cannot be discharged, and the power is purchased from the grid.

11) Case(11)

When $P_{dis,\min} < P_L < P_{dis,\max}$, $P_{dis,\min} < P_{dis,soc} < P_{dis,\max}$, and $P_L < P_{dis,soc}$, the energy storage can be discharged. Since $P_{dis,soc} < P_L$, the energy storage will discharge at $P_{dis,soc}$ until it is unable to discharge further, and the remaining power is purchased from the grid.

12) Case(12)

When $P_{dis,\min} < P_L < P_{dis,\max}$, $P_{dis,\min} < P_{dis,soc} < P_{dis,\max}$, and $P_L < P_{dis,soc}$, the energy storage can be discharged. Since $P_L < P_{dis,soc}$, the energy storage will discharge at P_L, without fully depleting, and no power is purchased from the grid.

13) Case(13)

When $P_{dis,\min} < P_L < P_{dis,\max}$ and $P_{dis,\max} < P_{dis,soc}$, the energy storage can be discharged. Because $P_{dis,\max} < P_{dis,soc}$, the energy storage will discharge at P_L, without fully depleting, and no power is purchased from the grid.

14) Case(14)

When $P_{dis,\max} < P_L$ and $P_{dis,soc} < P_{dis,\min}$, the energy storage cannot be discharged, and the power is purchased from the grid.

15) Case(15)

When $P_{dis,\max} < P_L$ and $P_{dis,\min} < P_{dis,soc} < P_{dis,\max}$, the energy storage can be discharged. The storage discharges at $P_{dis,soc}$ until it can no longer discharge, and the remaining power is purchased from the grid.

16) Case(16)

When $P_{dis,\max} < P_L$ and $P_{dis,\max} < P_{dis,soc}$, the energy storage can be discharged. However, due to P_L and $P_{dis,soc} > P_{dis,\max}$, the storage discharges at $P_{dis,\max}$ but will not be fully depleted, and the missing power is purchased from the grid.

VI. PSO-SA OPTIMIZATION ALGORITHM

The PSO–SA algorithm combines non-dominated sorting, crowding distance, and an external archive mechanism for application in multi-objective optimization. During iterations, particles update their velocities and positions, and suboptimal solutions are accepted with a certain probability according to the Metropolis criterion. The process continues until convergence or the maximum number of iterations is reached, with temperature decay applied in each iteration.

VII. CASE STUDY ANALYSIS

A. BASIC DATA

This study uses a specific data center for the case analysis object and extracts task data over a 12 day period. Considering the redundancy required for batch task scheduling, the total study duration is set to 12.5 days, with a sampling interval of 5 minutes. Fig. 8 illustrates the variations in wind speed

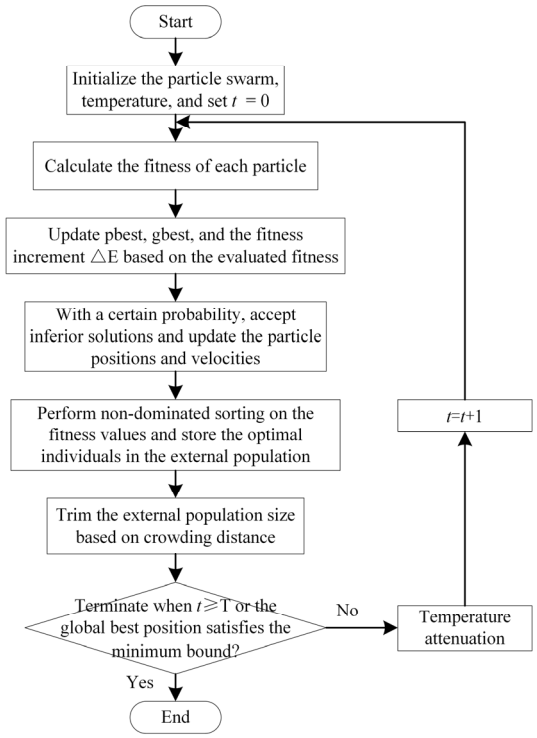


FIGURE 7. Flowchart of the PSO–SA optimization algorithm.

and solar irradiance during this period. The results indicate that wind speeds exceed 5 m/s for most of the time, suggesting abundant wind resources. Meanwhile, solar energy is also sufficiently available during daytime hours. Overall, the region demonstrates favorable wind and solar resource conditions, providing a solid foundation for the construction of a data center microgrid system.

The electricity price for the microgrid of the data center to purchase power from the main grid adopts a stepped pricing mechanism. The off-peak period is from 23:00 to 07:00, and the electricity price is 0.20 CNY/ (kWh). The peak hours are from 07:00 to 09:00, 14:00 to 18:00, and 21:00 to 23:00, with an electricity price of 0.98 CNY/ (kWh). The rest are during regular hours, with an electricity price of 0.51 CNY/ (kWh). The feed-in tariff for selling electricity back to the grid is uniformly set at 0.15 CNY/(kWh). Additional relevant parameters are detailed in Table 3.

The characteristics of partial workloads in the data center are illustrated in Table 4. The data center operates with a sampling interval of 5 minutes, continuously recording the task start time, computational demand, and delay tolerance. As shown in Table 3, tasks with zero delay tolerance are categorized as interactive loads, which must be executed immediately. In contrast, tasks with allowable delays ranging from 1 to 12 hours are classified as batch processing loads and must be completed within their respective deadlines. The temporal distribution of tasks across different time slots is depicted in Fig. 10. It can be observed that the data center experiences peak processing pressure during two critical periods: 07:25–10:50 and 12:45–15:20.

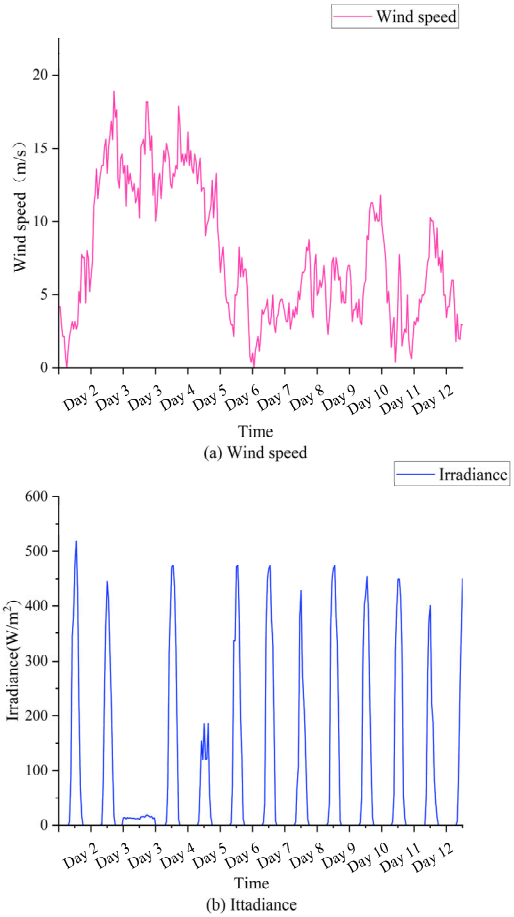


FIGURE 8. Wind speed and irradiance parameters.

TABLE 3. Main parameter setting of system.

Parameter	Value	Parameter	Value
v_{ci}	2m/s	Y_w	20years
v_r	12m/s	r_0	0.05
v_{co}	24m/s	c_{pv}	2400CNY/kW
G_r	1000W/m^2	c_{pv}^{mai}	0.03CNY/kWh
PUE	1.3	c_f^{pv}	0.0782kg CO_2/kWh
P_{idle}	0.15kW	Y_{pv}	20years
P_{peak}	0.3kW	r_1	0.05
c_w	2800CNY/kW	c_e^{lns}	2000CNY/kWh
c_{wt}^{mai}	0.05CNY/kWh	c_e^{mai}	0.018CNY/kWh
c_f^{wt}	0.1064kg gCO_2/kWh	c_f^g	0.4819kg CO_2/kWh

To alleviate the power demand during these high-load periods and enhance the utilization of renewable energy, a portion of

TABLE 4. Workload task volume (portion).

Start Time	Task Volume	Delay Tolerance (h)
11:15	544	4
11:20	6528	7
11:30	272	6
11:30	544	10
11:35	544	0

the batch tasks should be rescheduled to time slots with higher wind/solar generation or lower electricity prices. This enables a dual-objective optimization of load shifting and renewable energy absorption.

B. SENSITIVITY ANALYSIS

Fig. 9 shows that all three objectives remain within stable ranges. Construction cost and carbon emissions fluctuate due to periods of abundant wind and solar generation, during which the microgrid sells excess electricity to the grid to reduce costs and simultaneously lowers carbon emissions by decreasing electricity purchases.

As shown in Fig. 10, the 15th percentile of user satisfaction is 0.724, while the 5th-percentile CVaR is 0.689, indicating only a slight drop under extreme conditions and demonstrating controllable risk. When the satisfaction threshold exceeds 0.75, the probability of achieving high satisfaction increases sharply, indicating that the proposed method can maintain high user satisfaction across most scenarios.

Fig. 11 shows experimental groups with PUE $=$ 1.1, PUE $=$ 1.5, $\pm$ 15% renewable output, and the baseline. The proposed method demonstrates robustness in cost, carbon emissions, and user satisfaction: cost rises $\leq$ 5.12%, emissions $\leq$ 21.5%, and satisfaction fluctuates $\leq$ 28.86% with increasing PUE. $\pm$ 15% variations in renewable output result in minor changes, confirming adaptability to generation uncertainties.

Fig. 12 presents Monte Carlo simulations (1,000 runs) comparing scheduled and unscheduled strategies, where the unscheduled strategy processes tasks immediately upon arrival. The scheduling strategy reduces construction cost by 87.82%–97.99%, maintaining high reductions across all percentiles with a slight decreasing trend at higher percentiles, demonstrating a significant and stable optimization effect. Carbon emissions decrease by 46.26%–93.24%, also exhibiting a slight decline across percentiles while remaining substantial, highlighting the effectiveness of scheduling in lowering emissions.

C. ANALYSIS OF EXPERIMENTAL RESULTS FOR THE MULTIVARIATE COLLABORATIVE OPTIMIZATION SCHEDULING STRATEGY

This group integrates the first WSETSS scheduling strategy into the coordinated multi-objective optimization framework

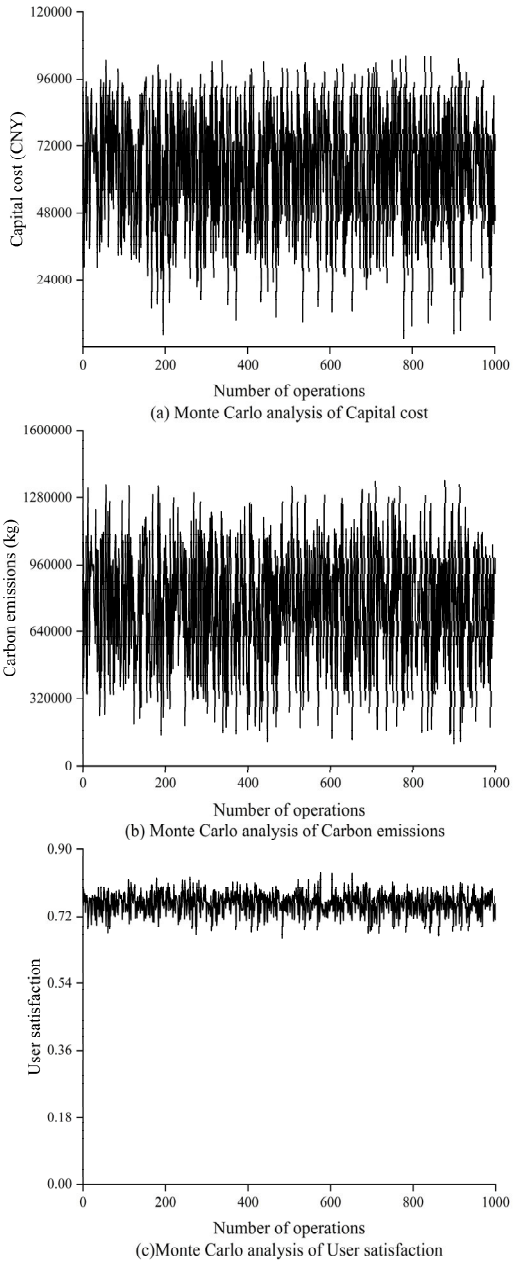


FIGURE 9. Monte carlo analysis of the three objectives.

of the data center microgrid to improve task allocation by leveraging renewable generation, electricity pricing, and time flexibility. The corresponding Pareto front, obtained through the hybrid Particle Swarm Optimization and Simulated Annealing (PSO–SA) algorithm, is presented in Fig. 13. In the figure, the red dots represent the set of Pareto-optimal solutions, whereas the blue dot denotes the optimal solution selected using the TOPSIS method [33].

To further investigate the trade-offs among different optimization objectives, pairwise comparisons were conducted for the objective functions in this experimental group. Dominated solutions were excluded, and only non-dominated Pareto solutions were retained, as illustrated in Fig. 14.

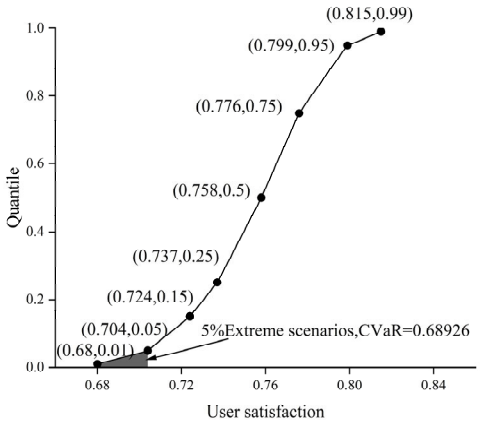


FIGURE 10. CVaR curve of user satisfaction.

Fig. 14(a) shows the relationship between construction cost and carbon emissions. A clear inverse trend is observed: as construction cost increases, carbon emissions decrease. This is because higher investments typically lead to greater installed capacities of wind and photovoltaic (PV) systems, thereby enhancing the utilization of renewable energy and reducing reliance on grid electricity, which in turn lowers carbon emissions. Notably, the fifth Pareto solution, compared to the fourth, involved an additional investment of CNY 4,037 in wind power capacity, which improved wind–solar output and resulted in a reduction of 92,937 kg in carbon emissions. This corresponds to a 4.56% increase in construction cost and a 12.07% reduction in carbon emissions. These findings indicate that moderate increases in renewable energy capacity can significantly contribute to carbon reduction.

Fig.14(b) depicts the relationship between construction cost and user satisfaction, which demonstrates a generally positive correlation. Increased investment in wind and PV capacity enhances renewable power availability. Under the scheduling strategy adopted in the first stage of Group 2, tasks are allocated to periods with high wind–solar output, improving the likelihood of timely execution and thus increasing user satisfaction. Further analysis reveals that when the construction cost ranges between CNY 80,000 and CNY 83,000, satisfaction improves significantly with increasing investment. However, beyond CNY 83,000, the improvement becomes marginal, showing a trend of diminishing returns. Therefore, within this investment range, moderate additional input into renewable energy capacity is an effective means of enhancing user satisfaction.

Fig. 14(c) explores the relationship between carbon emissions and user satisfaction. Interestingly, an increase in carbon emissions is, to some extent, associated with higher user satisfaction. This can be attributed to the two-stage scheduling strategy adopted in Group 2: the first stage prioritizes scheduling tasks during periods of peak renewable output, while the second stage follows a lowest-price-first

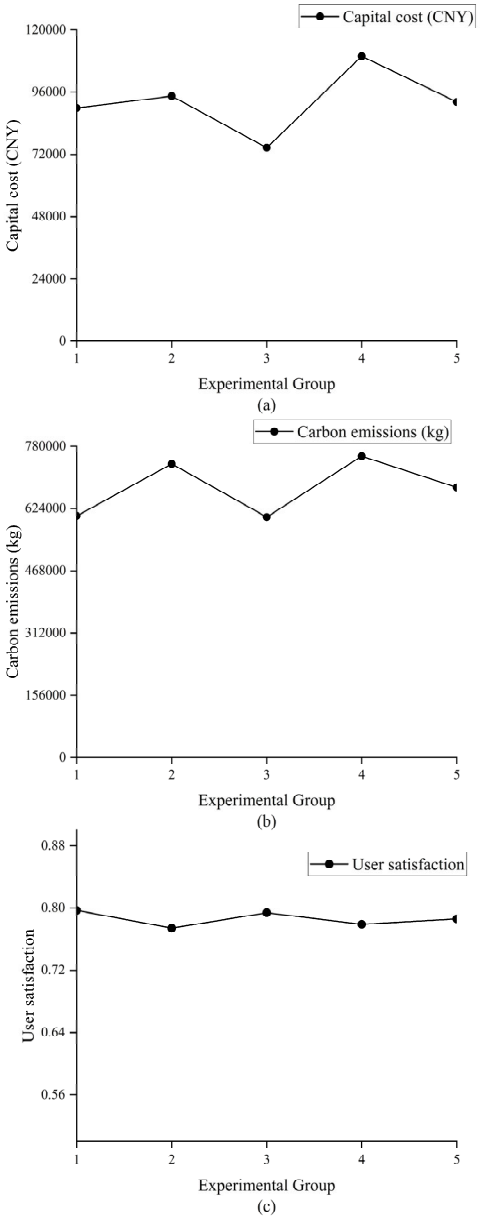


FIGURE 11. Validation under different operational condition.

principle. As illustrated in Fig. 10, wind power generation is concentrated in the early morning (00:00–03:00) and from 14:00 to 24:00, whereas PV generation peaks from 08:00 to 18:00. Overall, renewable output is skewed toward the latter half of the day. In the first stage, to match renewable peaks, tasks may be delayed, leading to lower satisfaction. In contrast, during the second stage, due to relatively stable electricity prices within the delay window, tasks are more likely to be scheduled closer to their arrival times, thereby improving satisfaction. As such, higher carbon emissions indirectly indicate a greater number of tasks being scheduled in the second stage, thus enhancing overall user satisfaction [34], [35].

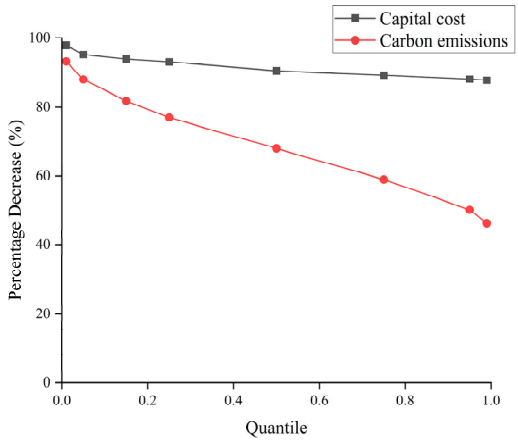


FIGURE 12. Comparison of scheduled vs. Unscheduled.

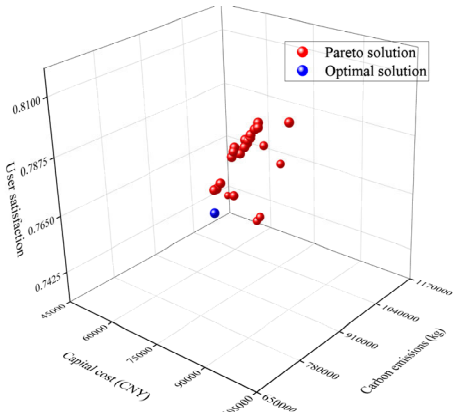


FIGURE 13. Pareto frontier results of multi-objective optimization.

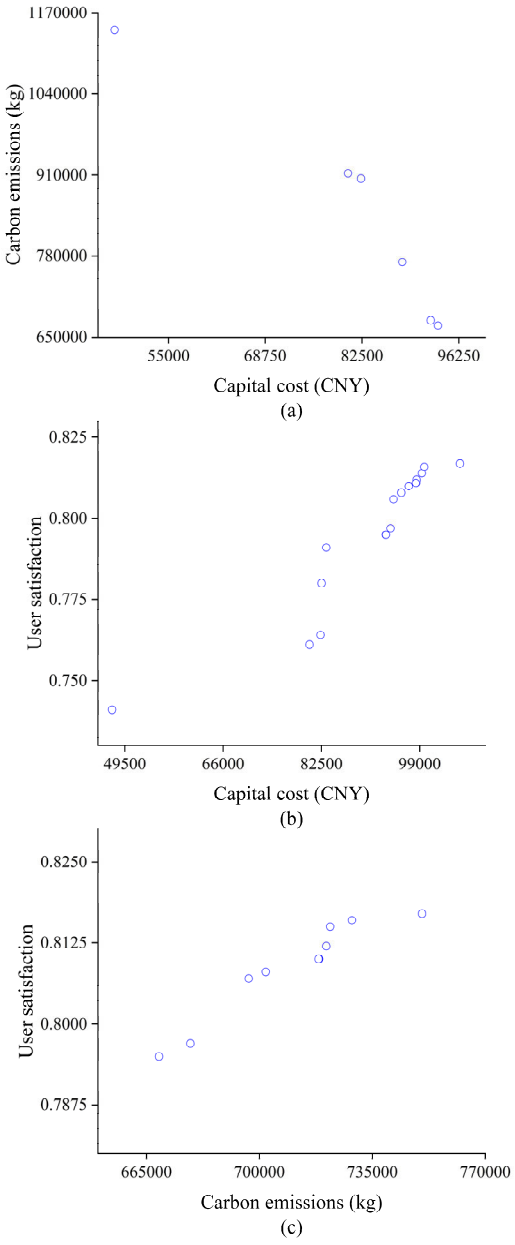


FIGURE 14. The two-dimensional unfolding diagram of the optimization result.

To enhance the applicability of the proposed scheduling strategy to real-world production environments, this study defines 13 distinct task delay tolerance windows, and task data were simulated using a 5-minute sampling interval [36]. This approach enables the data center to perform more flexible and dynamic scheduling based on the delay tolerance of each task, thereby further exploiting the temporal flexibility of workloads and improving resource allocation efficiency.

As illustrated in Figs. 15 and 16, the proposed strategy significantly improves the temporal distribution of task execution. A maximum processing limit of 10,000 tasks per time slot was imposed. When the task volume exceeds this threshold at a given time, the scheduling system redistributes tasks across adjacent time slots to achieve load leveling. Prior to scheduling, several time slots exhibited task demand far beyond processing capacity. After scheduling, the workload was evenly distributed, with no slot exceeding the upper limit, resulting in effective load balancing.

During the early stages of task arrival (before 15:20), tasks were mainly divided into two categories. The second category includes tasks with long maximum delay tolerance. Since the first stage of the second experiment prioritizes scheduling during periods of high renewable generation, these tasks were concentrated in the 07:25–15:20 time window—characterized by peak wind and solar output. This period originally experienced low workload; scheduling tasks here not only improved renewable energy utilization but also enabled near-full-load operation, significantly enhancing resource use efficiency. The second category consists of tasks arriving between 00:00–07:00 with short delay tolerance. Given the limited renewable generation and low task flexibility during this period, these tasks bypass the first stage and are scheduled in the second stage based on the lowest electricity price principle. Since this window aligns with the

off-peak tariff (CNY 0.20/kWh), these tasks are scheduled close to their arrival times, balancing timeliness, electricity cost, and user satisfaction [37].

For tasks arriving after 15:20, the system primarily schedules them into three time periods: (1) 16:20–19:50, where solar output decreases but wind power increases rapidly; (2) 20:20–22:00, characterized by low solar but peak wind generation; (3) After 23:00, when renewable output is low but electricity price enters an off-peak period. Periods (1) and (2) are associated with secondary renewable peaks and are prioritized by the first-stage strategy. Period (3), despite low renewable output, is advantageous in cost. To prevent resource congestion from last-minute task execution before daily deadlines, the strategy does not enforce completion by 24:00. Instead, it allows tasks to be postponed to the next day, as long as their maximum delay constraint is met, especially targeting periods with stronger renewable output. Some tasks constrained by tight delay windows are still executed in the second stage based on the lowest-price-first principle, typically around 23:00. Note that Figures 13 and 14 only show results for one day (288 time slots); deferred tasks scheduled for the following day are not included in the visualization.

In summary, the proposed scheduling strategy couples renewable generation profiles, electricity pricing trends, and task delay tolerance to establish a two-stage, category-based temporal reallocation mechanism. This enables the system to simultaneously optimize energy efficiency and operational economics, while maintaining acceptable levels of user satisfaction.

As shown in Fig. 17, three renewable generation peaks occur between 07:50 and 22:55. After scheduling, the data center's load curve closely aligns with the renewable output profile [38]. During 00:00–04:05 and 23:00–24:00, tasks constrained by tight delay tolerance are scheduled according to the second-stage low-price strategy, leading to execution during off-peak pricing. In contrast, for the 04:10–07:45 interval, when renewable generation is zero, the system relies on energy previously stored from excess wind generation to complete a small volume of task execution, rather than forcing them into high-renewable periods.

Overall, the scheduling results closely align with the expected behavior in the second group of experiments, validating the effectiveness of the proposed strategy. By considering task time constraints, the approach achieves coordinated optimization of renewable energy utilization and system operational cost. It dynamically adapts to renewable fluctuations and electricity price variations, enabling smart temporal task allocation, improving renewable energy penetration, and reducing dependency on the grid. The strategy ensures task deadlines are met while maintaining a balance between clean energy use and economic efficiency, demonstrating strong optimization performance and practical potential for engineering applications. It offers a valuable scheduling paradigm for constructing green and efficient data center energy management systems.

D. COMPARATIVE ANALYSIS OF THE FIVE WSETSS EXPERIMENTAL GROUPS

To evaluate the impact of different scheduling strategies on the operational performance of a data center microgrid, this study embeds five *WSETSS*-based task scheduling approaches into the proposed coordinated multi-objective optimization framework, designing five comparative experimental groups. Specifically:Group 1 adopts a strategy that prioritizes maximum renewable output and minimum electricity price; Group 2 schedules tasks based on the chronological order of renewable output combined with minimum electricity price; Group 3 uses maximum renewable output and minimum task completion time as the scheduling criterion; Group 4 is based on the chronological order of renewable output and minimum task completion time; Group 5 relies solely on the minimum electricity price for task scheduling. These experimental designs aim to explore the trade-offs between renewable energy utilization, electricity cost, and task execution performance under different scheduling priorities.

Fig. 18 compares the TOPSIS-selected optimal solutions derived from the hybrid Particle Swarm Optimization and Simulated Annealing algorithm across the five experimental groups. Overall, each group exhibits distinct trade-offs among construction cost, carbon emissions, and user satisfaction. From the perspective of construction cost, Group 4 achieved the lowest investment, primarily due to its comparatively lower installed capacities of wind, PV, and energy storage—especially a substantially reduced PV capacity—thereby minimizing the initial capital expenditure.

In terms of carbon emissions, Group 4's limited renewable energy utilization, caused by its relatively small wind and PV capacities, resulted in higher emissions. Although Group 5 adopted larger renewable capacities, its scheduling strategy prioritized time slots with the lowest electricity prices rather than those with high renewable output. As a result, the reliance on grid electricity remained substantial, and carbon emissions were not significantly reduced.

Regarding user satisfaction, the differences in scheduling strategies played a critical role. In Groups 1 and 3, tasks were scheduled during the first stage to periods with high renewable generation. However, since most renewable output is concentrated in the latter half of the day, some tasks experienced delayed processing, leading to lower user satisfaction. In contrast, Groups 2 and 4 adopted a sequential allocation of renewable resources based on chronological order, which reduced task delays and notably improved user satisfaction. Group 5, guided solely by the principle of lowest electricity prices, experienced slight reductions in user satisfaction compared to Groups 2 and 4. This was because a small portion of tasks was postponed to later time slots under the tiered electricity pricing mechanism. Nevertheless, as most tasks were still processed near their arrival times, the

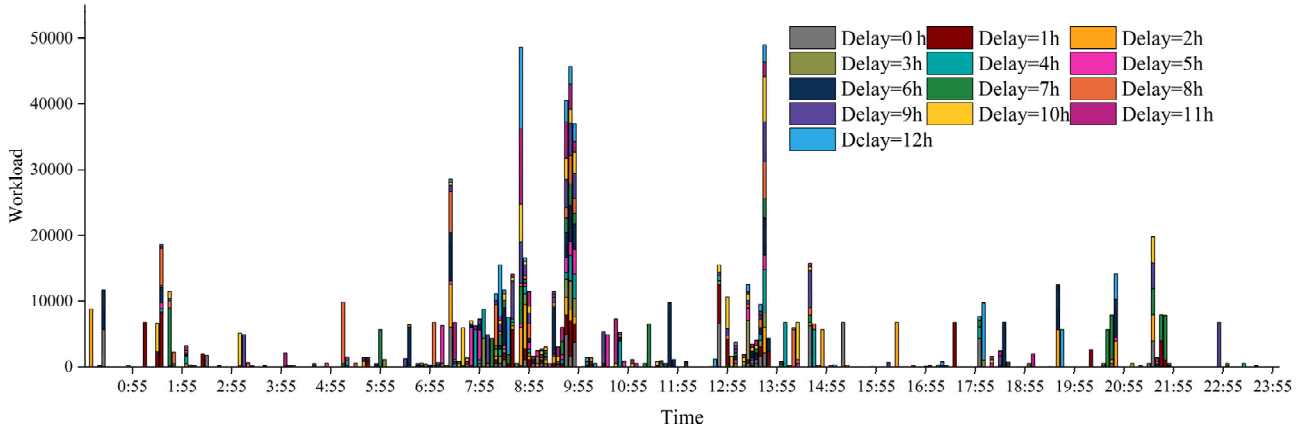


FIGURE 15. Task distribution before data center scheduling.

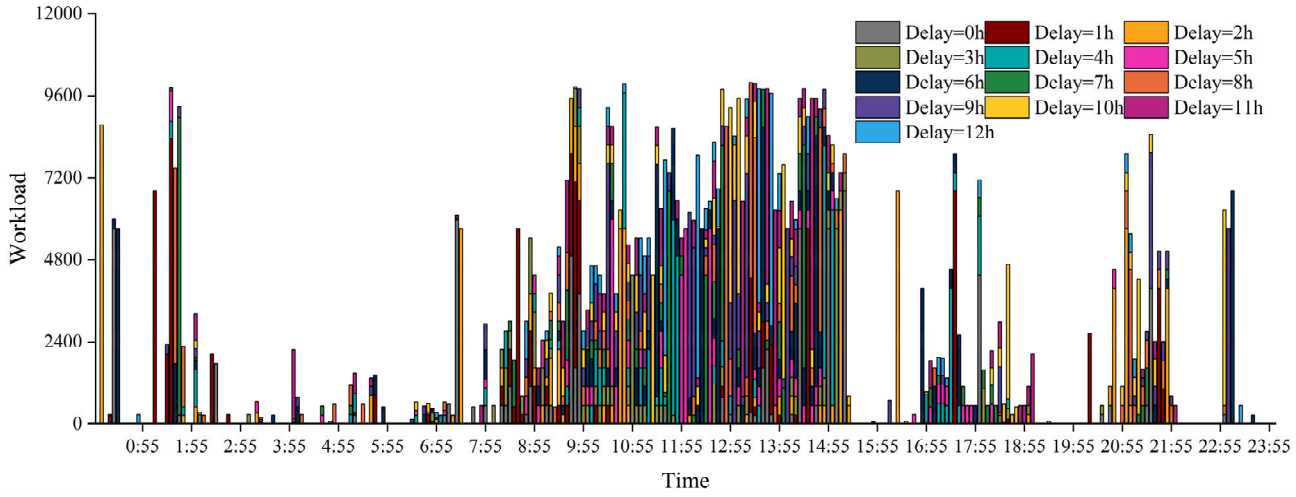


FIGURE 16. Task distribution after data center scheduling.

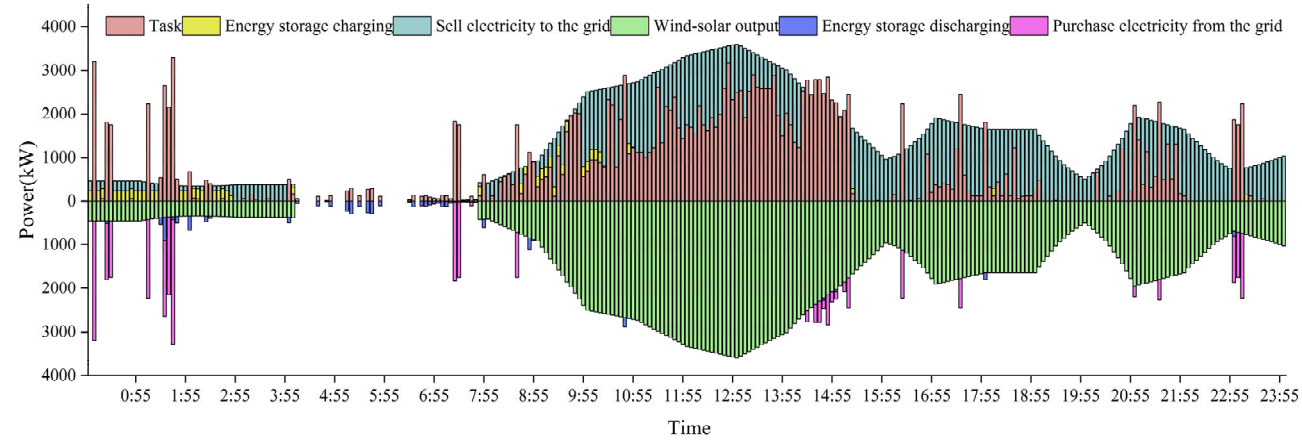


FIGURE 17. Power balance at all times in the data center.

satisfaction levels in Group 5 remained higher than those in Groups 1 and 3.

Additionally, the corresponding configurations of wind, PV, and energy storage capacities for each group are provided

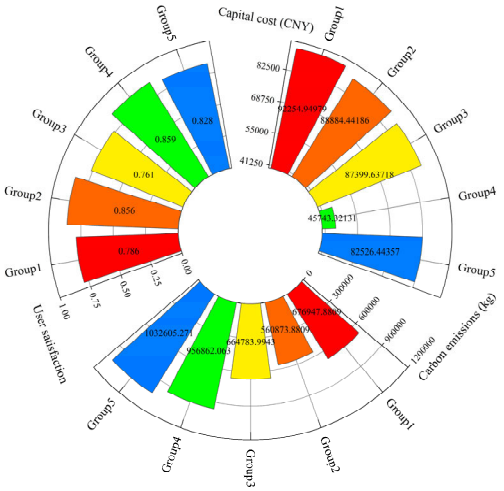


FIGURE 18. Comparison of the results of five groups of experiments.

TABLE 5. Optimal configuration results.

Group	Wind Capacity	PV Capacity	ES Capacity
Group 1	5000kW	5000kW	290kWh
Group 2	5000kW	5000kW	0kWh
Group 3	5000kW	4675kW	170kWh
Group 4	3845kW	1450kW	145kWh
Group 5	4190kW	4670kW	275kWh

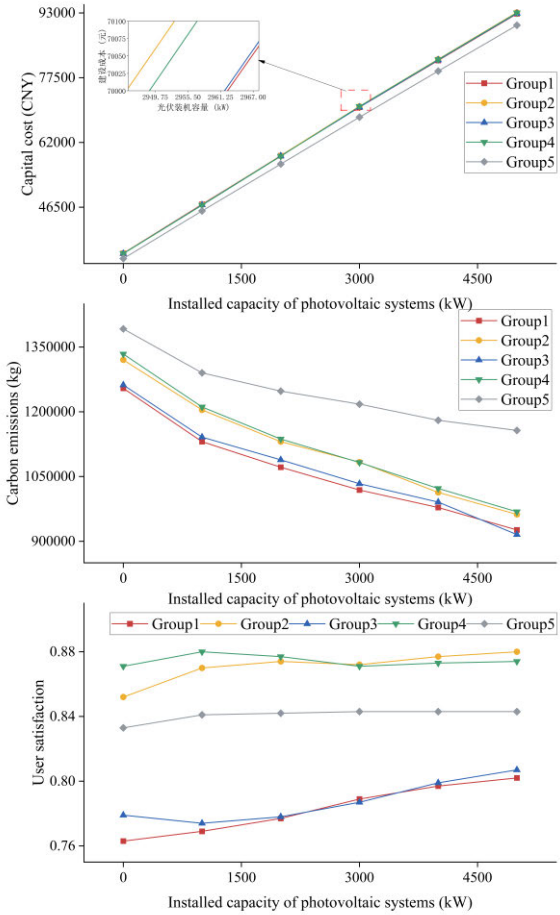


FIGURE 19. Photovoltaic and energy storage remain unchanged, but the wind turbines have changed.

in Table 5, offering a foundation for further analysis of the relationship between renewable energy configurations and system performance under different scheduling strategies.

Fig. 19 illustrates the impact of varying wind turbine installed capacity on system objectives under fixed PV capacity (3,000 kW) and energy storage capacity (1,000 kWh).

In terms of construction cost, an overall increase is observed across all experimental groups as wind capacity increases, primarily due to the additional investment in wind power infrastructure. However, Group 5 consistently exhibits the lowest construction cost. This is attributed to its scheduling strategy, which allocates tasks to the lowest electricity price periods while simultaneously selling excess renewable generation back to the grid. As a result, total system cost is reduced by approximately 2.92% compared to other groups.

Regarding carbon emissions, a general downward trend is evident with increasing wind capacity across all groups, verifying the decarbonization potential of renewable energy in offsetting grid electricity. Enhanced wind output strengthens system reliance on clean energy, thereby significantly reducing indirect emissions. Notably, Groups 1 and 3, which prioritize scheduling tasks during periods of peak renewable generation in the first stage, achieve superior emission reductions—3.99% to 23.09% lower than the other three groups [39]. This improvement stems from their ability to significantly boost local renewable utilization and minimize dependence on external electricity supply [40]. In contrast, despite possessing relatively large installed renewable capacity, Group 5 fails to fully utilize available green power. Because its scheduling strategy focuses solely on electricity price, many tasks are allocated to low-price periods with limited renewable availability, resulting in higher grid dependency and only marginal emission reductions. This highlights the need for co-optimization between resource allocation and scheduling strategies to achieve meaningful decarbonization. Moreover, Group 4 shows significantly higher emissions when wind capacity is set to 0 kW, underscoring the fact that scheduling optimization alone cannot compensate for insufficient renewable resource capacity [41]. Therefore, effective low-carbon operation depends on the dual synergy between renewable configuration and scheduling mechanisms—ensuring rational allocation of wind and solar resources while guiding load consumption toward clean energy through well-designed scheduling strategies.

In terms of user satisfaction, when wind capacity is 0kW and the system relies solely on PV, the renewable supply is

limited. Under such conditions, the scheduling strategies in Groups 1 through 4 prioritize renewable utilization in their first stage, resulting in task deferral and lower satisfaction levels. In contrast, Group 5, which schedules tasks based on the lowest electricity price without renewable output constraints, can allocate tasks closer to their arrival times, thereby improving user satisfaction by 3.02% to 20.93% compared to the other groups. Further comparison reveals that under identical first-stage strategies, Groups 3 and 4, which adopt minimum completion time in the second stage, deliver higher satisfaction than Groups 1 and 2, which rely on the lowest electricity price. This is because minimum completion time scheduling better ensures timely task execution. Additionally, Group 1, which schedules tasks exclusively at renewable peaks, suffers from the longer wait times needed to reach those peak periods, resulting in the lowest overall user satisfaction.

As wind capacity increases beyond 1,000 kW, the number of high renewable output time slots expands, reducing the need for task delay. Under these conditions, Groups 2 and 4, which adopt chronological consumption of renewable output in the first stage, not only schedule tasks closer to their arrival times but also make effective use of clean energy. Consequently, their user satisfaction gradually surpasses that of Group 5, with a maximum increase of 21.29%. This finding demonstrates that under high renewable penetration, scheduling strategies that account for both resource availability and task response characteristics can simultaneously improve system efficiency and significantly enhance user satisfaction [42].

As the variation in PV capacity under fixed wind (3,000 kW) and energy storage (1,000 kWh) conditions yields system performance trends similar to those shown in Fig. 19, the corresponding figure is not presented here.

VIII. CONCLUSION

The coordinated multi-objective optimization scheduling strategy proposed in this study effectively achieves peak shaving and load leveling, significantly enhancing the alignment between load demand and renewable generation. This leads to a substantial improvement in clean energy utilization and reduces grid dependence.

The five *WSETSS*-based comparative experiments reveal a strong synergy between renewable energy configuration and scheduling strategies. Even under identical wind and solar capacities, different scheduling approaches lead to significant performance differences. Price-oriented strategies reduce total system cost by 2.92%, while renewables-prioritized strategies lower carbon emissions by 3.99%–23.09%. User satisfaction improves by up to 21.29% when scheduling better aligns with renewable generation under high wind capacity. These findings highlight that optimal system performance requires more than high renewable penetration or a single scheduling logic—coordinated optimization of both energy allocation and scheduling is key to balancing cost, emissions, and service quality in green data centers.

Overall, the proposed framework enables low-carbon, high-efficiency operation in green data centers. Future work will focus on incorporating renewable uncertainty, integrating dynamic task patterns from real-world Google and Facebook datasets, extending WSETSS with reinforcement learning, and including market-based electricity pricing to further enhance robustness and efficiency.

APPENDIX A
SUPPLEMENTARY EQUATIONS

$$
\begin{cases}
C_1 = \dfrac{\theta_w c_w P_{wr}}{365} T \\
\theta_w = \dfrac{r_0 (1+r_0)^{Y_w}}{(1+r_0)^{Y_w} - 1} \\
C_2 = \dfrac{\theta_{pv} c_{pv} P_{pvr}}{365} T \\
\theta_{pv} = \dfrac{r_1 (1+r_1)^{Y_{pv}}}{(1+r_1)^{Y_{pv}} - 1} \\
C_3 = \dfrac{\theta_e c_e^{\ln s} y_e}{365} T \\
\theta_e = \dfrac{r_2 (1+r_2)^{Y_e}}{(1+r_2)^{Y_e} - 1} \\
C_4 = c_{wt}^{mai} P_{wr} T \\
C_5 = c_{pv}^{mai} P_{pvr} T \\
C_6 = y_e c_e^{mai} T \\
C_7 = P_1 P_{line}^t \Delta T \\
C_8 = p_2 p_{ex}^t \Delta T \\
S_i = 1 - \dfrac{T_{com}}{T_{del}} \\
F_i = \begin{cases} 1, T_{com} \le T_{del} \\ -1, T_{com} > T_{del} \end{cases}
\end{cases}
$$

APPENDIX B
WSETSS PSEUDOCODE

Step 1: If $P_{mom} > P_t + S_{oc}^t$ or the computing resources are insufficient, proceed to Step 2.

Step 2: Obtain the following parameters within the delay-tolerant time window T_{del}:

Renewable power output: P_{ws}

Electricity price: P_{price}

Available computing resources: Res

Step 3: Select the scheduling strategy:

Strategy 1 → execute Step 4

Strategy 2 → execute Step 5

Strategy 3 → execute Step 4

Strategy 4 → execute Step 5

Strategy 5 → execute Step 6

Step 4: Sort P_{ws} in descending order within T_{del}, and traverse the sorted time slots:

If $P_{ws} > P_{mom}$ and computing resources are sufficient, schedule the task;

Else:

If using Strategy 1 → go to Step 6

If using Strategy 3 → go to Step 7

Step 5: Traverse time slots within T_{del} in chronological order:

If $P_{ws} > P_{mom}$ and computing resources are sufficient, schedule the task;

Else:

If using Strategy 2 → go to Step 6

If using Strategy 4 → go to Step 7

Step 6: Sort P_{price} in ascending order within T_{del}, and traverse the sorted time slots:

If computing resources are sufficient, schedule the task;

Else → go to Step 8

Step 7: Traverse time slots within T_{del} in chronological order:

If computing resources are sufficient, schedule the task;

Else → go to Step 8

Step 8: Schedule the task sequentially in the time slots outside T_{del}, then proceed to Step 9.

Step 9: Return to the main scheduling framework to process the next task or update the system state.

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